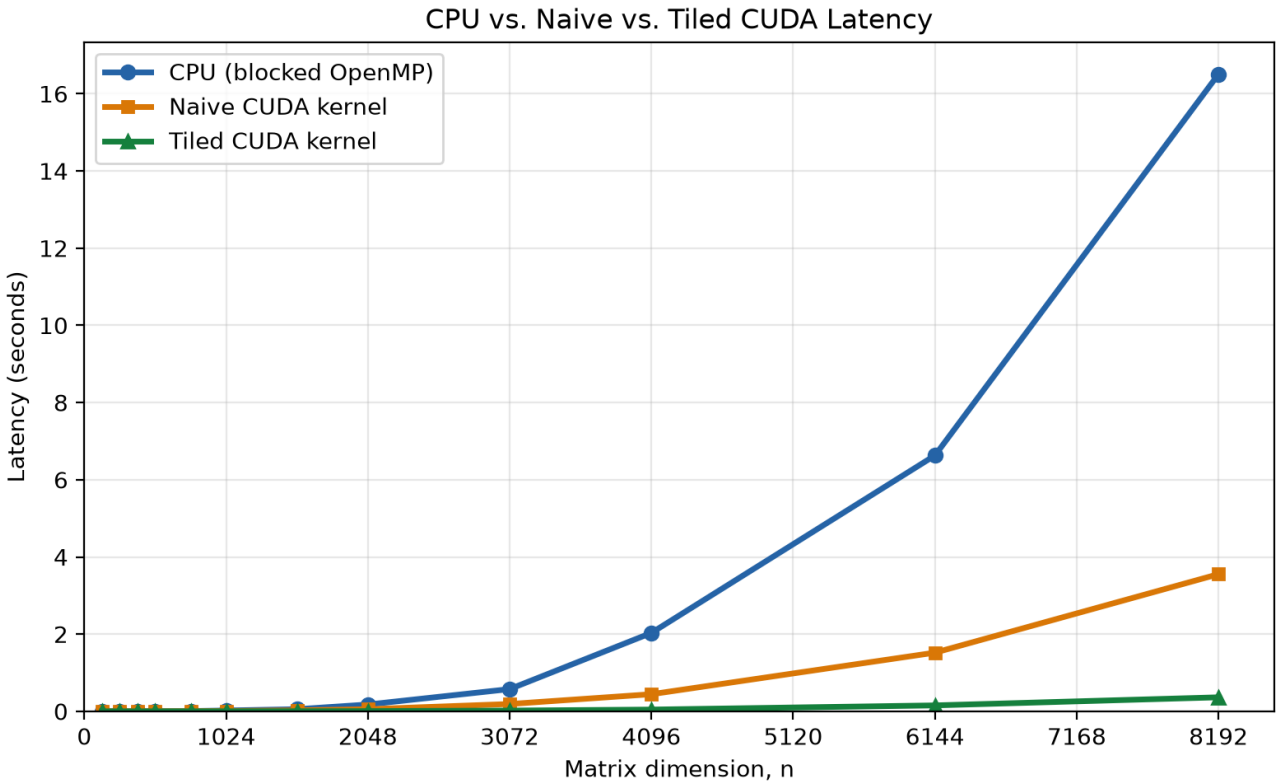
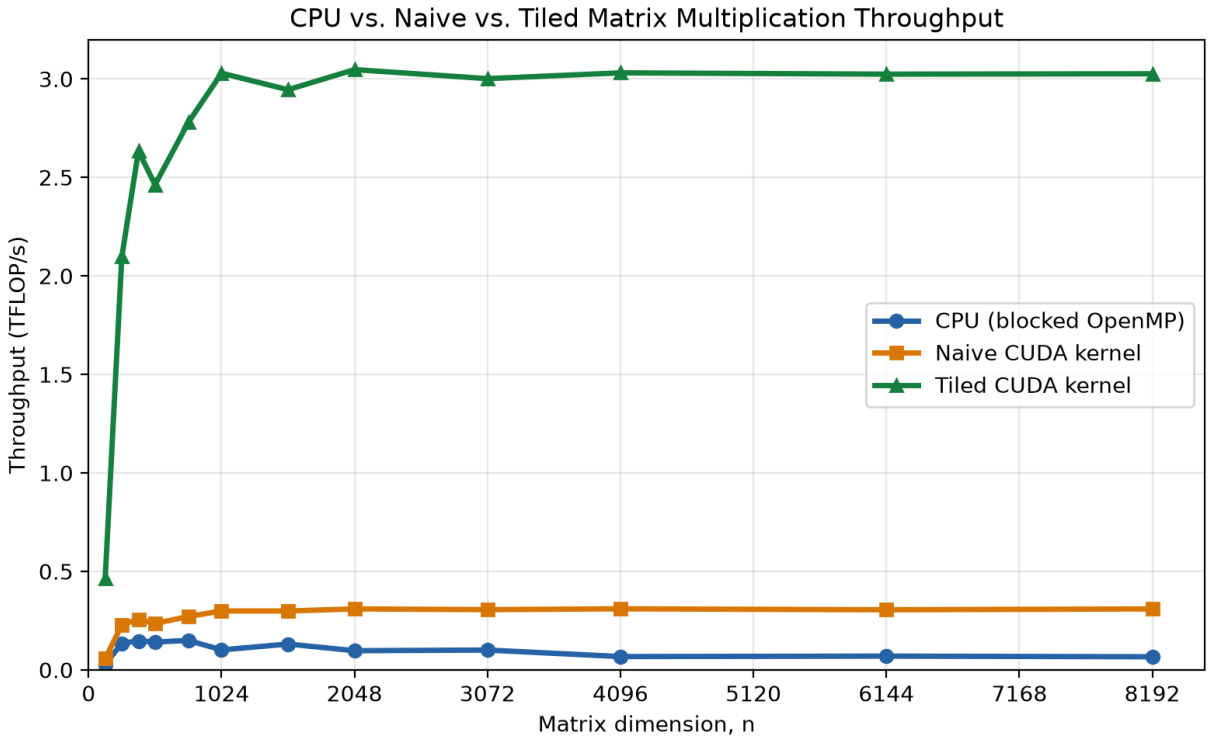
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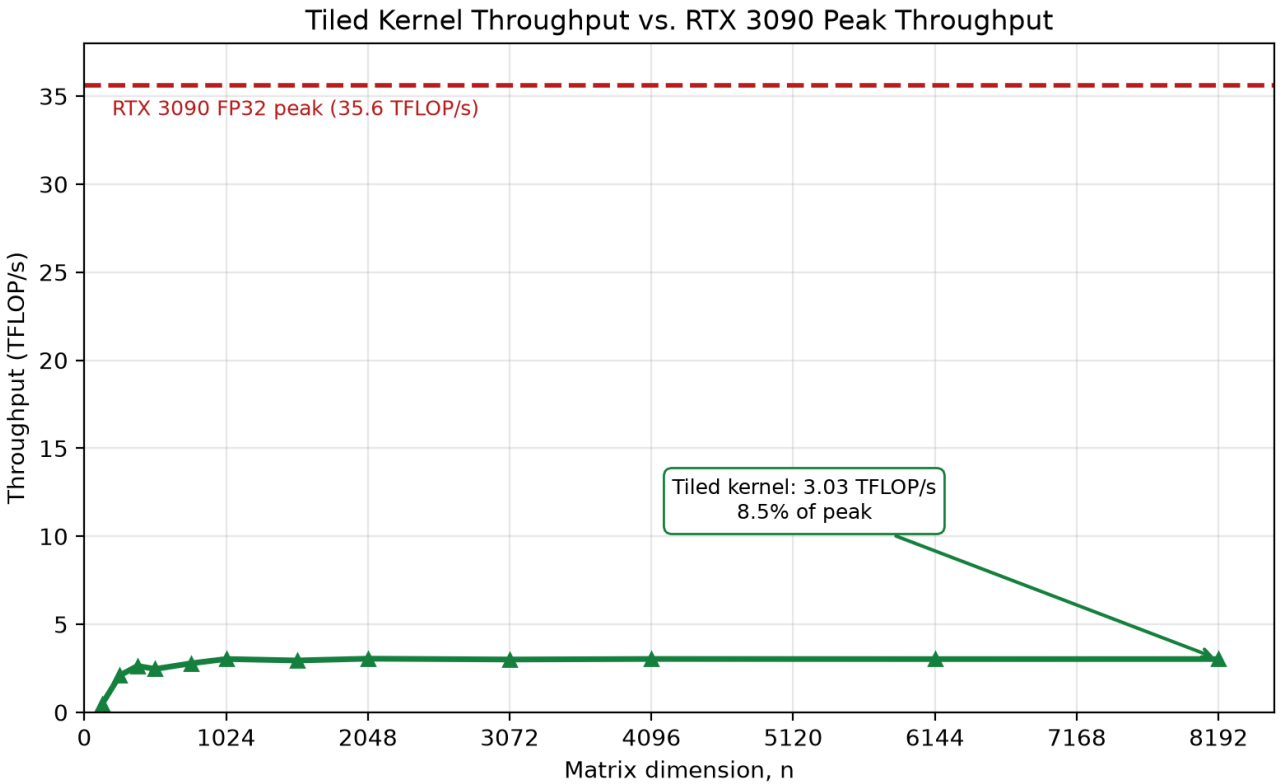
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Why throughput

Throughput tells us

1. A general performance metric, no matter the problem size
2. How well we are utilizing hardware

Sidenote: why throughput varies with matrix size

We just talked about why throughput is a better performance because it is normalized against problem size. However, the plots clearly show that *throughput* itself varies with problem size!

Take throughput at large problem size

The achieved throughput is still calculated from measured latency. At small problem sizes, overhead is a larger portion of measured latency, artificially deflating the achieved throughput. Thus, we tend to take the throughput at the largest reasonable problem size. Also notice that throughput stabilizes at larger problem sizes.

Throughput as a performance measure

Throughput

Throughput is the total useful work (FLOPs) done per unit time. *We previously discussed theoretical peak throughput of hardware.*

$$\text{Throughput} = \frac{\text{FLOPs}}{\text{time}}$$

Example

Peak throughput of our RTX 3090 is 35.6 TFLOP/s. We observed 360ms or 0.36s of latency for $n = 8192$, where the total FLOPs required is 1.10 TFLOPs. Thus, we have an observed throughput of

$$\frac{1.10\text{TFLOPs}}{0.36\text{s}} \approx 3.05\text{TFLOP/s}.$$

Roughly 8.5% of peak throughput.

Arithmetic Intensity

Arithmetic Intensity

We define the **Arithmetic Intensity** (AI) of our kernel as the number of FLOPs done per bytes moved from DRAM.

$$\text{AI} = \frac{\text{FLOPs}}{\text{bytes moved}}$$

Example

For $n, k, m = 8192$, our tiled kernel moved a total of 137.7 GB of data from DRAM. With a total work count of 1.10 TFLOP, its arithmetic intensity is

$$\text{AI} = \frac{1.10 \times 10^{12} \text{ FLOP}}{137.7 \times 10^{9} \text{ bytes}} \approx \boxed{8.0 \text{ FLOP/byte}}.$$

Roofline Model

Roofline Model

For a given system, we can predict a the **upper bound** of throughput of a kernel given its AI. Up to the **ridge point**, we are **memory bound**, increasing AI will raise the upper bound of throughput. However, at the **ridge point**, we become **compute bound**, and no longer benefit from increasing AI.

Example

For our 3090, we have 35.6 TFLOP/s of peak FP32 throughput and 936 GB/s of memory bandwidth.

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At AI $= 8.0$ FLOP/byte, we are below the ridge point, and are still memory bound.

Roofline Model of our RTX 3090

